

HW9: Tracing with references

1.

Statement	Variable Values
	m → 143 k → 70
m.transferFrom(k);	
	m → 70 k → 0

2. Because the transferFrom method need to change the value of m and k.

If there is an immutable type, the value cannot be changed.

3.

1.

Statement	Variable Values
private static void swap1(int i1, int i2) {	
	i1 = 7 i2 = 12
int tmp = i1;	
	i1 = 7 i2 = 12 tmp = 7
i1 = i2;	
	i1 = 12 i2 = 12 tmp = 7
i2 = tmp;	
	i1 = 12 i2 = 7 tmp = 7
}	
Start tracing here	
int x = 7, y = 12;	
	x = 7 y = 12
swap1(x, y);	
	x = 7 y = 12

2.

Statement	Variable Values
<code>private static void swap2(String s1, String s2) {</code>	
	s1 → legends s2 → leaders
<code>String tmp = s1;</code>	
	s1, tmp → legends s2 → leaders
<code>s1 = s2;</code>	
	s1, s2 → leaders tmp → legends
<code>s2 = tmp;</code>	
	s1 → leaders s2, tmp → legends
<code>}</code>	
Start tracing here	
<code>String x = "legends", y = "leaders";</code>	
	x → legends y → leaders
<code>swap2(x, y);</code>	
	x → legends y → leaders

3.

Statement	Variable Values
<code>private static void swap3(NaturalNumber n1, NaturalNumber n2) {</code>	
	n1 → 41 n2 → 78
<code>NaturalNumber tmp = n1;</code>	
	n1, tmp → 41 n2 → 78
<code>n1 = n2;</code>	
	n1, n2 → 78 tmp → 41
<code>n2 = tmp;</code>	
	n1 → 41 n2, tmp → 78
<code>}</code>	
Start tracing here	
<code>NaturalNumber x = new NaturalNumber2(41), y = new NaturalNumber2(78);</code>	
	x → 41 y → 78
<code>swap3(x, y);</code>	
	x → 41 y → 78